

Subjective and Objective Analyses of Collaboration and Co-Presence in a Virtual Reality Remote Environment

Allison Bayro*

Yalda Ghasemi†

Heejin Jeong‡

University of Illinois at Chicago

ABSTRACT

Remote collaboration in virtual reality has gained attention and proved to be a viable solution for providing effective collaboration environments for physically distant collaborators. This study compares head-mounted display (HMD)- and computer-based remote collaboration solutions that allow users to interact with each other through immersive environments. Analyzing remote collaboration in immersive environments requires understanding group interactions and personal experiences. For this purpose, a 3D object assembly task was performed by 10 participants using self-reported surveys and physiological measures to investigate the effectiveness of collaboration from the users' perspective. The results showed that the HMD-based remote collaboration in a virtual reality environment increased the sense of co-presence among the users.

Keywords: Virtual environment, virtual reality, co-presence, remote collaboration, physiology.

Index Terms: Human-centered computing [Human computer interaction (HCI)]: Interaction paradigms—Mixed/augmented reality; Human-centered computing [Human computer interaction (HCI)]: Interaction paradigms—Collaborative interaction; Human-centered computing [Collaborative and social Computing]: Collaborative and social computing theory, concept and paradigms—Computer supported cooperative work

1 INTRODUCTION

Collaboration is found across many areas including industrial, medical, and educational domains [1, 6, 14]. Collaborative scenarios have advanced from collocated tasks to increasingly more advanced, remote efforts worldwide, especially in the COVID-19 pandemic. The commonly used technological interfaces, such as Zoom, Skype, Facebook, enable remote collaboration and support conceptualizing, designing, and prototyping solutions for enhanced communication, cooperation, training, and learning [9]. However, researchers report the drawback of using 2D interfaces as a collaborative environment, given collaborators' lack of shared space [10].

Extended reality (XR) technologies have been used in many applications and domains from manufacturing and human-robot collaboration to more daily routine tasks such as office tasks [3, 11, 15]. Further, XR technologies are able to support tasks involving more than one user. XR was developed to facilitate shared spaces to enable remote collaboration. Recent studies using XR remote collaboration technologies have evaluated features, such as virtual annotations and gestures to promote shared understandings amongst collaborators [7].

*Co-first author, e-mail: abayro2@uic.edu

†Co-first author, e-mail: zghase3@uic.edu

Allison Bayro and Yalda Ghasemi contributed equally and they are co-first authors.

‡Corresponding author, e-mail: heejinj@uic.edu

Other studies aimed at developing and analyzing XR platforms to enable an operator-worker scenario of remote collaboration where one collaborator (e.g., a remote operator) views and instructs the other collaborator (e.g., a local worker) on the steps to perform a task [8, 12].

Many researchers have used physiological measurements to provide an objective and non-invasive way to evaluate human behavior and cognitive states in virtual reality (VR). Dey et al. (2018) explored how providing heart rate or respiration rate as audio-haptic feedback between collaborators influenced collaboration [4]. The results found that real-time heart rate feedback increased collaborators' presence and emotional understanding. Salminen et al. (2019) investigated shared visualization of physiological signals in social VR to evoke feelings of empathy between collaborators [13]. The respiration rate was visualized as a pulsating aura around avatars and empathy-related motivation from the frontal asymmetry of electroencephalography (EEG) data was portrayed as a color cue. The results showed that frontal asymmetry was more effective in evoking empathy than respiration biofeedback. According to [5], while brain-related signals such as EEG contribute most to the sense of spatial presence and co-presence, other physiological measurements (e.g., heart rate and skin conductance responses) are considered as the secondary effects of presence. However, these secondary measurements have been less investigated in the previous studies especially in terms of skin galvanic responses or electrodermal (EDA) activity.

Limited studies investigated VR remote collaboration platforms where both individuals contribute *equally* to a task. There is not enough research on how these collaborative platforms compare while using a VR device or a standard 2D interface, as well as physiological stress and arousal responses to these platforms. Thus, there is a need to evaluate both subjective and objective measures of collaboration in a shared environment where both individuals can equally and simultaneously contribute to the task. In this study, we hypothesized that VR collaborative environments increase the spatial presence and co-presence among the users while increasing arousal.

2 METHODS

2.1 Participants

A total of 10 participants (i.e., five pairs including 7 males and 3 females) with an ages of 22 to 27 ($M = 24.3$, $SD = 1.64$) were recruited. They were mostly unfamiliar and inexperienced in AR/VR collaborative platforms while being highly familiar with traditional remote collaboration platforms, such as Zoom and Skype.

2.2 Apparatus

This study used a commercial mixed reality-based remote collaboration platform, *Spatial*TM, to provide a collaborative remote environment. Two Oculus Quest 2 Virtual Reality HMD headsets and two desktop monitors with a mouse and a keyboard were used to enable remote collaboration. Participants were instructed to work together to assemble the provided models (Figure 1) as quickly and accurately as possible. Figure 2 shows 3D object assembly tasks from the perspectives of (a) HMD and (b) PC users.

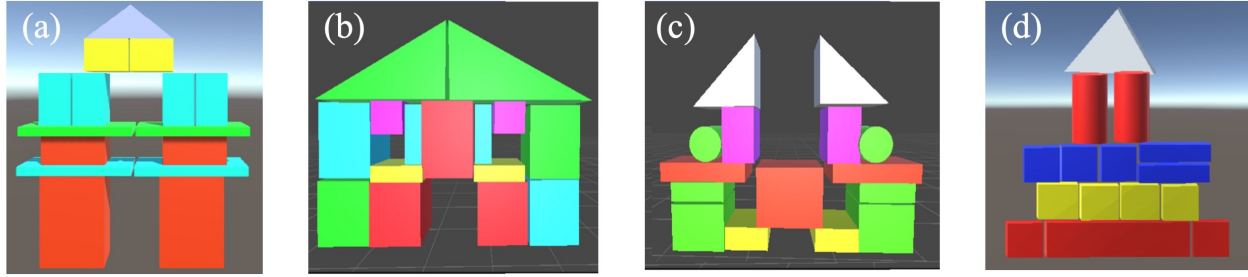


Figure 1: Four 15-piece models (a) skyscraper, (b) house, (c) cat face, (d) rocketship.

2.3 Experimental Design

The goal of this study is to evaluate the collaboration of two remote users for a 3D object assembly task in the form of constructing a model from its 15 component pieces in four different modes: 1) both subjects worked remotely via a non-immersive platform, such as a computer (PC-PC), 2) Subject 1 worked remotely via a computer while Subject 2 worked remotely via an HMD Oculus 2 VR Device (PC-HMD), 3) Subject 2 worked remotely via a computer while Subject 1 worked remotely via HMD, and 4) both subjects worked remotely via HMD VR Devices (HMD-HMD). The experiment was a counterbalanced within-subject design, and each participant experienced all task conditions. Although HMD-PC and PC-HMD can be considered as the same conditions, remote collaboration through Spatial platform offers different experiences on HMD and PC, as the experience in PC is more limited compared to immersive VR experience of HMD. Therefore, we evaluated HMD-PC and PC-HMD separately and these two conditions were not used interchangeably. Namely, the HMD-PC condition takes into account the subjective and objective results of the participants on the HMD platform while their partner is on the PC platform. Conversely, the PC-HMD condition considers the results of the participants on the PC platform while their partner is on the HMD platform. The experience of each user and their collaborator is different in these cases.



Figure 2: 3D object assembly task scenes from (a) HMD and (b) PC users.

Despite being in different locations, the participants were able to communicate verbally with each other through an integrated

microphone/speaker on the HMD and/or computer. The task completion time was recorded during the experiment. After completion, they completed igroup Presence Questionnaire (IPQ).

Physiological Sensor Placement and Filtering: For each experimental pair, one subject was prepped for physiological data collection from electrodermal activity (EDA) sensors (BIOPAC Systems, Inc.). For the EDA data collection, the palm of the non-dominant hand was cleaned and a small amount of isotonic gel was applied on the electrode and the electrodes were placed on the cleaned locations.

All subsequent analyses were performed using Biopac AcqKnowledge software. A median smoothing filter was applied to the signal to remove minor motion artifacts [2]. Subsequently, the signal was resampled at 50Hz, and the phasic component was extracted. This signal component provides information on the fast changes associated with short-term events and occurs in discrete environmental stimuli (i.e., sensory stimuli and cognitive processes). Phasic changes show as abrupt increases, called "peaks" [2]. The data for each experimental session was standardized from baseline data for each subject. The number of non-specific skin conductance responses (NS-SCRs) per minute was calculated and averaged for each session.

3 RESULTS

igroup Presence Questionnaire (IPQ): For the *General sense of presence*, *Spatial Presence*, and *Involvement (or Co-Presence)*, HMD-HMD and HMD-PC modes were rated significantly higher than the other modes. In terms of *Realism*, no significant difference was observed (Figure 3).

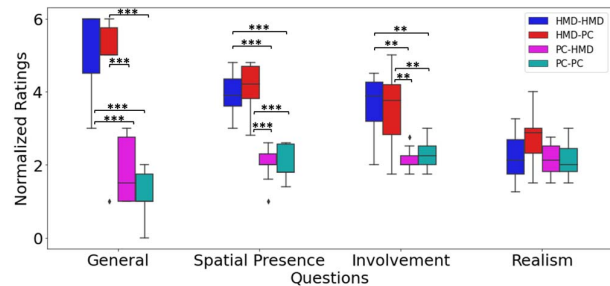


Figure 3: The results of IPQ questionnaire.

Electrodermal Activity (EDA): For the NS-SCRs, the results showed significantly more peaks per minute in the HMD-HMD than the PC-HMD and PC-PC, showing higher arousal which can be an indication of a higher sense of co-presence. Despite not finding significant differences in the other pairs, upon visual inspection, we see the HMD-HMD and HMD-PC sessions have higher average values for peaks per minute than the PC-HMD and PC-PC (Figure 4).

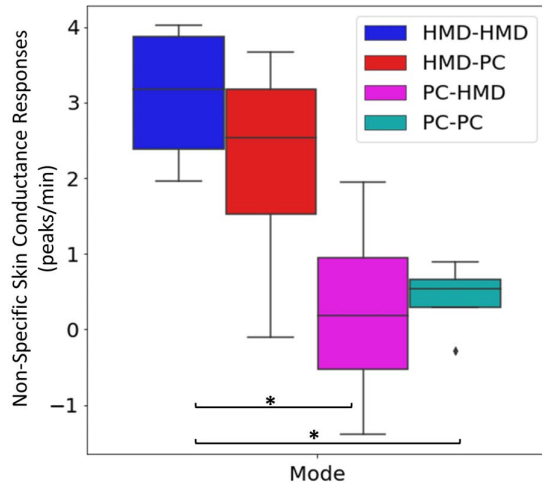


Figure 4: NS-SCR (peaks/min) for the four collaborative modes.

4 CONCLUSION, LIMITATIONS & FUTURE DIRECTIONS

In this study, we utilized physiological sensors to measure skin conductance to validate the results of a subjective self-reported survey. Participants reported a higher sense of presence in VR. The output of EDA also confirmed this result. It can be concluded that VR collaborative environments using a HMD increase the sense of collaboration including spatial presence and co-presence. For small-scale use cases that do not require engagement and contribution from all attendees, video conferencing is considered the most efficient type of collaboration. However, for organizing events with many participants requiring individuals' engagement, XR-based collaborative platforms are increasingly taking over. In case of geographic limitations, immersive environments for collaboration are even more effective than in-person meetings. However, there are also some limitations when working on these platforms. When using 3D spatial spaces on computers, the users cannot see the depth as they see in an immersive virtual space and cannot easily manipulate the objects. This makes it difficult to compare with VR immersive environments that offer more options and freedom for interactions. Users acknowledged that VR immersive environments can be even superior to traditional platforms. Using HMDs for collaboration increased the sense of spatial presence among the users, which can be a great advantage of remote collaboration using VR. However, HMDs need more improvement in their design to be more efficient for long-term use. Also, this study will be further investigated with more participants and other physiological measurements such as heart rate. With this result, it can be concluded that AR/VR immersive environments are a great alternative to overcome the limitations of remote working while engaging people through modern technologies.

ACKNOWLEDGMENTS

This work was supported in part by the National Science Foundation Graduate Research Fellowship Program awarded by A.B.

REFERENCES

- [1] L. Alem and J. Li. A study of gestures in a video-mediated collaborative assembly task. *Advances in Human-Computer Interaction*, 2011, 2011.
- [2] J. J. Braithwaite, D. G. Watson, R. Jones, and M. Rowe. A guide for analysing electrodermal activity (eda) & skin conductance responses (scrs) for psychological experiments. *Psychophysiology*, 49(1):1017–1034, 2013.
- [3] S. H. Choi, K.-B. Park, D. H. Roh, J. Y. Lee, M. Mohammed, Y. Ghasemi, and H. Jeong. An integrated mixed reality system for safety-aware human-robot collaboration using deep learning and digital twin generation. *Robotics and Computer-Integrated Manufacturing*, 73:102258, 2022.

- [4] A. Dey, H. Chen, C. Zhuang, M. Billinghurst, and R. W. Lindeman. Effects of sharing real-time multi-sensory heart rate feedback in different immersive collaborative virtual environments. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 165–173. IEEE, 2018.
- [5] S. Grassini and K. Laumann. Questionnaire measures and physiological correlates of presence: A systematic review. *Frontiers in Psychology*, 11:349, 2020.
- [6] S. Johnson, M. Gibson, and B. Mutlu. Handheld or handsfree? remote collaboration via lightweight head-mounted displays and handheld devices. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing*, pp. 1825–1836, 2015.
- [7] S. Kim, M. Billinghurst, and K. Kim. Multimodal interfaces and communication cues for remote collaboration. *Journal on Multimodal User Interfaces*, pp. 1–7, 2020.
- [8] G. A. Lee, T. Teo, S. Kim, and M. Billinghurst. A user study on mr remote collaboration using live 360 video. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 153–164. IEEE, 2018.
- [9] B. Marques, A. Teixeira, S. Silva, J. Alves, P. Dias, and B. S. Santos. A critical analysis on remote collaboration mediated by augmented reality: Making a case for improved characterization and evaluation of the collaborative process. *Computers & Graphics*, 2021.
- [10] W. Matcha and D. R. A. Rambli. Exploratory study on collaborative interaction through the use of augmented reality in science learning. *Procedia Computer Science*, 25:144–153, 2013.
- [11] K.-B. Park, S. H. Choi, J. Y. Lee, Y. Ghasemi, M. Mohammed, and H. Jeong. Hands-free human-robot interaction using multimodal gestures and deep learning in wearable mixed reality. *IEEE Access*, 9:55448–55464, 2021.
- [12] T. Rhee, S. Thompson, D. Medeiros, R. Dos Anjos, and A. Chalmers. Augmented virtual teleportation for high-fidelity telecollaboration. *IEEE Transactions on Visualization and Computer Graphics*, 26(5):1923–1933, 2020.
- [13] M. Salminen, S. Järvelä, A. Ruonala, V. Harjunen, G. Jacucci, J. Hamari, and N. Ravaja. Evoking physiological synchrony and empathy using social vr with biofeedback. *IEEE Transactions on Affective Computing*, 2019.
- [14] M. Schneider, J. Rambach, and D. Stricker. Augmented reality based on edge computing using the example of remote live support. In *2017 IEEE International Conference on Industrial Technology (ICIT)*, pp. 1277–1282. IEEE, 2017.
- [15] A. Singh, Y. Ghasemi, H. Jeong, M. Kim, and A. Johnson. A comparative evaluation of the wearable augmented reality-based data presentation interface and traditional methods for data entry tasks. *International Journal of Industrial Ergonomics*, 86:103190, 2021.